DEPARTMENT OF MATHEMATICS
UNIVERSITY OF CAPE COAST
MAT 101: ALGEBRA AND TRIGONOMETRY QUIZ II

ATTEMPT ALL QUESTIONS. **TIME ALLOWED: 50 MINS**

NAME:..... **INDEX NUMBER:**.....

Circle the most appropriate choice.

1. Find the oblique asymptote for the function $C(t) = \frac{t^3+50}{3t^2+t}$, $t \geq 0$.

   A. $C = 3t - 9$ B. $C = 0$

   C. $C = \frac{1}{3}t - \frac{1}{9}$ D. No oblique asymptote

2. Determine the vertex of the equation $y = (x+3)^2-5$ which is a translation of the parent function $y = x^2$.

   A. $(3, 5)$ B. $(-3, -5)$ C. $(-5, 3)$ D. $(5, -3)$

3. Which equation will shift the graph of $y = x^2$ left by 5 units and up by 6 units?

   A. $y = (x + 6)^2 - 5$ B. $y = (x + 5)^2 + 6$

   C. $y = (x + 5)^2 - 6$ D. $y = (x - 5)^2 + 6$

4. What is the fourth term of the expansion of $(3x - 2)^5$?

   A. $-240x$ B. $-720x^2$ C. $720x^2$ D. $1080x^3$

5. For the rational function $f(x) = \frac{-4x^2+18x+16}{x^2+8x+7}$, find all vertical and horizontal asymptotes.

   A. vertical asymptotes $x = 1$ and $x = -7$, horizontal asymptote $y = -4$.

   B. vertical asymptotes $x = -1$ and $x = -7$, horizontal asymptote $y = -1/4$.

   C. no vertical asymptotes, horizontal asymptote $y = -4$.

   D. vertical asymptotes $x = -1$ and $x = -7$, horizontal asymptote $y = -4$.

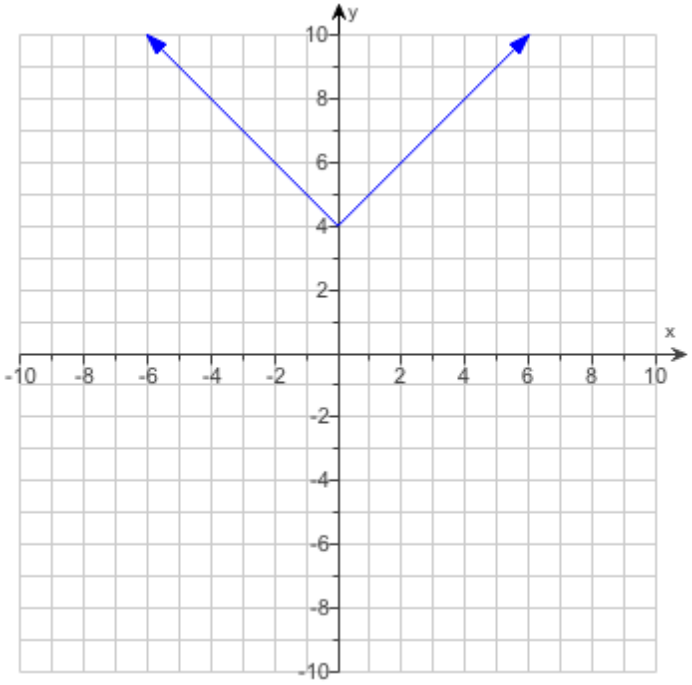
6. The fourth term in the expansion of $(x-2y)^{12}$ is:

A. $-1760x^9y^3$
B. $-7160x^9y^3$
C. $-1670x^9y^3$
D. $-1607x^9y^3$

7. Which of the following functions has no horizontal asymptote?

A. $f(x)=\frac{x^2-2}{x^3-9x-4}$
B. $f(x)=\frac{5x^4-2x^3-x+7}{-x^3+3x^2-4}$
C. $f(x)=\frac{9x^2-2x-3}{x^2+8x-2}$
D. $f(x)=\frac{1}{x^2-x}$

8. What is the equation of the absolute value function?



A. $f(x)=|x+4|$
B. $f(x)=|x|+4$
C. $f(x)=|x-4|$
D. $f(x)=|x|-4$

9. Let $f(x)$ be a function and $q>1$. Which of the following is a true statement?

A. $-f(x)$ is a reflection of $f(x)$ in y-axis
B. $f(\frac{x}{q})$ is a shrink in y-direction by factor q
C. $f(qx)$ is a translation of $f(x)$, q units to the left.
D. $\frac{1}{q}f(x)$ is a shrink in y-direction by factor q

10. Given that $f(x) = \frac{x^2+1}{x^2-3}$. Determine the domain and the range of $f(x)$.

A. Domain = $\{x : x \in \mathbb{R},$ except $x = \pm\sqrt{3}\}$ and range = $(-\infty, -\frac{1}{3}] \cup (1, \infty)$

B. Domain = $\{x : x \in \mathbb{R},\ x \neq \pm\sqrt{3}\}$ and range = $(-\infty, -\frac{1}{3}) \cup [1, \infty)$

C. Domain = $\{x : x \in \mathbb{R},\ x \neq \pm\sqrt{3}\}$ and range = $(-\infty, \frac{1}{3}) \cup [1, \infty)$

D. Domain = $\{x : x \in \mathbb{R},\ x \neq \pm\sqrt{3}\}$ and range = $(-\infty, \frac{1}{3}] \cup (1, \infty)$

11. Which of the following is a rational function with an x-intercept of $(-1, 0)$, a hole at $x = 3$ and a vertical asymptote at $x = 4$?

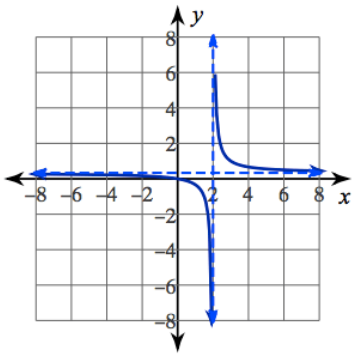
A. $y = \frac{(x-1)(x+3)}{(x+4)(x+3)}$ B. $y = \frac{(x+1)(x-3)}{(x-4)(x-3)}$

C. $y = \frac{(x-4)(x-3)}{(x+1)(x-3)}$ D. $y = \frac{(x+4)(x+3)}{(x+1)(x+3)}$

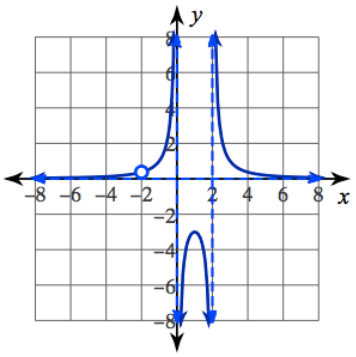
12. Choose the correct graph that matches with the equation.

$$f(x) = \frac{3x - 6}{x}$$

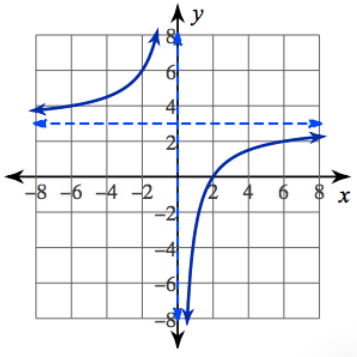
A.



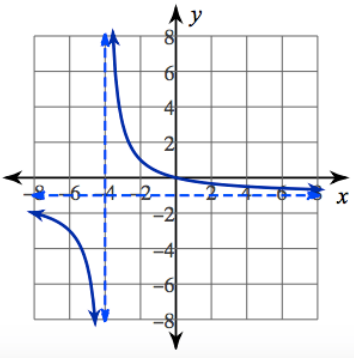
B.



C.



D.



13. If $\log_{27} x = \frac{1}{6}$, find x.

A. $\sqrt{3}$ B. 3 C. 9 D. 27

14. Simplify $\log(\frac{4lm}{k})$.

A. $2\log 2 + \log l + \log m - \log k$ B. $\log 2 + \log l - \log m + \log k$

C. $\log 4 - \log l - \log m - \log k$ D. $\log 4 + \log l + \log m + \log k$

15. Determine the domain of the function $f(x) = \frac{4}{x^2+1}$.

A. $\{x : x \in \mathbb{R}\}$

B. $\{x : x \in \mathbb{R}, x \neq 1\}$

C. $\{x : x \in \mathbb{R}, x \neq -1\}$

D. $\{x : x \in \mathbb{R}, x \neq 4\}$

16. Which of the following asymptotes will the graph of the function $f(x) = \frac{(x-1)(x+2)}{(x+2)(x-3)}$ have?

A. Vertical and Horizontal Asymptotes

B. Vertical and Oblique Asymptotes

C. Oblique and Horizontal Asymptotes

D. No asymptotes

17. Find the 8th term of the sequence whose general term $U_n = 2n^2 - 3n + 1$.

A. 105 B. 110 C. 115 D. 120

18. What is the domain of the function $f(x) = x^{\frac{1}{2}}$?

A. $(2, \infty)$ B. $(-\infty, 1)$ C. $[0, \infty)$ D. $[1, 2)$

19. Find m if $\log_8 m + \log_8 \frac{1}{6} = \frac{2}{3}$.

A. 4 B. 12 C. 18 D. 24

20. Determine the values of G and H if $\frac{4x-5}{x^2-x-2} \equiv \frac{G}{x-2} + \frac{H}{x+1}$.

A. $G = 1, \ H = 3$ B. $G = 1, \ H = -3$

C. $G = 3, \ H = 1$ D. $G = 3, \ H = -1$